

A micro-meso-macro perspective on the methodology of evolutionary economics: Integrating history, simulation and econometrics

John Foster · Jason Potts

Abstract We develop a new methodology for evolutionary economic modelling, based upon the ‘micro-meso-macro’ analytical framework for dealing with behaviour in complex economic systems. This new methodology involves a combination of computational, historical and econometric methods. It is argued that this integrated methodology is superior to the theoretically restricted simulation/calibration methodology found in new classical macroeconomics and to the relatively unrestricted methodology found in much of agent-based modelling in evolutionary economics. It is also viewed as much more useful than the conventional ‘vector error correction’ econometric methodology when the evolutionary economic modeller is faced with non-linear time paths and associations between observed variables.

Keywords Micro-meso-macro · Evolutionary economic modelling · Agent based modelling · Economic methodology · Calibration

JEL Classification C15 · C51 · C52

1 Introduction

Applied economics has long been dominated by multiple regression techniques. Despite this, the simulation and calibration method has recently become viewed by some economists as an alternative to traditional

We would like to thank those who commented on an earlier draft of this article at the International J.A. Schumpeter Society Conference in Sophia-Antipolis, Nice, France, June 21–24, 2006.

J. Foster (✉)

School of Economics, University of Queensland, Australia
e-mail: j.foster@uq.edu.au

J. Potts

School of Economics, University of Queensland, Australia

econometric methods. However, in contrast to econometrics, there is not yet a well developed methodology to guide this kind of modelling, although some good progress has recently been made in this regard (see Brenner and Werker 2006).

In mainstream economics, simulation/calibration came to prominence in the 1980s in the real business cycle (RBC) literature (see Kydland and Prescott 1982, 1996) after it was discovered that an RBC model cannot be estimated econometrically because of the very strong and unrealistic assumptions that are made (Altug 1989). However, the calibration usually attempted is minimalist. It is argued that, if the variances of simulated and actual time series data are found to match in some proximate sense, this constitutes evidence that RBC is supported. This methodology has been heavily criticized, both within and beyond the new classical school of thought (see, for example, Hoover 1995). Unsurprisingly, many applied macroeconomists have also been unimpressed with RBC simulation/calibration, preferring to apply econometric modelling approaches that are built upon statistical foundations. Vector error (or equilibrium) correction (VEC) representations of time series data are overlaid by 'long run equilibrium' restrictions, suggested by economic theory and verified using co-integration analysis, have been preferred. Whereas RBC theorists are primarily concerned with broad explanation, VEC modellers are more interested in forecasting of particular macroeconomic variables.

In evolutionary economics, there has never been a strong tradition of using econometrics, beyond the investigation of special questions, such as the parametric structure of innovation diffusion curves. The reason for this is clear: when significant structural change is present, conventional econometric methods are mostly unsuitable for empirical research because the classical assumptions required for valid econometric analysis are breached. Also, the evolutionary economic perspective on microeconomics offers no coherent way that microeconomics and macroeconomics can be linked. Thus, the main focus has been on the behaviour of firms and the industries that they populate. In this context, simulation/calibration has been applied to explore the outcomes of economic processes within firms, between firms and between firms and consumers. Agent-based modelling (ABM) became increasingly popular following the seminal contribution of Nelson and Winter (1982). This has developed into a 'history friendly' methodology (see Malerba et al. 2001). The goal is to conduct ABM in contexts that, as accurately as possible, reflect the historical and institutional conditions that existed in the period of time under consideration. Thus, simulation/calibration is conducted in precisely the opposite context to RBC – it embraces history rather than eliminating it.

However, as Werker and Brenner (2004) point out, the ABM simulation/calibration approach faces a difficulty: it is possible to generate a wide range of ABM models that can calibrate on a given set of time series data since there are no formal restrictions that can be placed on all of the chosen parameters. This led them to argue for a critical realist methodology that,

somewhat like the history friendly approach,¹ involves considerable historical and case study investigation prior to simulation. This results in models that are quite specific to the firm or industry in question. The stylized representations of historical processes that are obtained are then used for counterfactual experiments, but it is uncommon for researchers in this tradition to draw out general theoretical principles from these simulation exercises that might, for example, provide an analytical basis for evolutionary macroeconomics in the style of, say, Metcalfe, Foster and Ramlogan (2006).

History friendly modelling is, essentially, about the existence and adoption of rules that guide economic behaviour. It belongs to what Nelson and Winter (1982) referred to as 'appreciative theory,' rather than 'formal theory.' Replicator dynamic simulations are preferred to formal analytical methods to understand the outcomes of economic behaviour in firms and industries. Inevitably, this results in a 'supply side' emphasis. The demand side, although discussed, does not receive as much attention (Foster and Potts 2006). Furthermore, although the importance of routines in firms is emphasized, much less attention is given to the generic rules that facilitate coordination in the wider economy (Dopfer and Potts 2007). It was not until the 1990s, that a literature began to develop in evolutionary economics where it was argued that, because the process of variety generation is often markedly non-random, generic rules have to be treated in a different way to those in classical genetics. Among others, Foster (1997, 2000) and Witt (1997) argued for an explicit self-organisational approach to dealing with the emergence of variety in evolutionary economics. Then, following Kirman (1983), Potts (2000) argued for a 'network and connections' system that could integrate competitive selection and self-organisation processes. This evolved into the 'micro-meso-macro' framework of Dopfer, Foster and Potts (2004) in which rule systems (and how they adapt) are the key building blocks in complex economic systems (see also Dopfer and Potts 2004, 2007).

The question that is addressed in the remainder of this article is: can we develop a methodology that can connect macroeconomics with microeconomics via the 'micro-meso-macro' analytical framework?

2 Micro-meso-macro

The micro-meso-macro framework places generic rules at the centre of economic analysis. The economic system is viewed as being made of cognitive, behavioural, socio-cultural, organizational, technical and institutional rules. The analytic concept of a meso unit is a rule and its population of carriers and, in this sense, the economy is made of meso units. Microeconomic analysis is the study of the individual carriers of rules and their local operations, and

¹ But see Brenner and Werker (2006) for a more fine-grained comparison of their approach to that of Malerba et al. (2001).

macroeconomic analysis is the study of the effects of coordination and change in the meso structure of the whole economy. From this perspective, economic evolution involves the origination, adoption and retention of a novel meso rule in the micro and macro structure of the economy.²

Economic activity relies upon the existence of embedded generic rules with stable carrier populations. These have been referred to variously in the past as institutions, norms, understandings, laws, technologies, etc. They are rules that are held by more than one person and are ‘fourth order complex’ in the sense that an individual believes that another upholds the rule and *vica versa* (see Foster 2005, Klochko and Ordeshook 2006). If this communality of understanding exists at the meso level in a carrier population, then economic activity can occur and value can be generated. The meso domain of analysis, then, is the foundation for both evolutionary micro and macroeconomic analysis.³

The evolutionary micro domain contains carriers engaged in a myriad of activities (or operations) facilitated by meso rules resulting in a heterogeneous range of processes and products that have economic value that can be aggregated up to a defined macroeconomic level. Some generic rules have long lives and operate at the core of the economic structures, such as the rules of markets, property rights, hierarchy or other forms of organization. Others are short lived, coming and going in fashions, such as the flared-trousers rule. At a point in time, economic activity is determined by its structure of meso rules, some of which are deep and stable, others of which are shallow and passing. Economic evolution arises from change in that generic structure. This is abstractly conceived as a three-phase rule-trajectory: (1) the origination of the novel rule as an innovation, (2) the adoption of that rule into a population of micro agents, and (3) the ongoing retention and use of the meso rule. New meso populations come into being when a previously idiosyncratic rule becomes the basis of an innovation and is adopted by other carriers because its application yields significant economic value.⁴ Old generic rules die out when their applications cease to be of value and their meso populations diminish.

Economic structures are incomplete networks of connections, which can be technological or organizational. Some connections are hard-wired and generic while others are relatively impermanent and peripheral. The operation of rules yields ‘work’ both in the traditional thermodynamic sense (technological rules) and in the organizational effectiveness sense (social rules). There is economic value if output is in demand. In turn, demand relies upon

² Dopfer and Potts (2007) outline this analytical framework from ontological first principles.

³ Note that this is in contrast to RBC models that seek micro, not meso, foundations for macro.

⁴ A patent is valuable if other carriers can use it profitably. So what is a technological rule is a saleable commodity rather than a meso rule. The meso rule in operation in such cases is the one that make patents possible in the mutual observation of patent (property) rights. However, such commodities can become meso rules if it becomes understood that a rule will be used generally, i.e., they will acquire significant public good characteristics. Microsoft Windows is a good example.

connections existing between products and preferences (Earl and Potts 2004). It is because organizational connections are so important both within and between economic entities, such as producers and consumers, that knowledge is so important. Every marketing manager knows that information must be provided in order to establish product knowledge amongst consumers, every production manager knows that shared knowledge is essential to make an organization function and every entrepreneur knows that a viable business must make these connections.

Economic structures, like all dissipative structures, must be open to absorb free energy. But they must also be open to absorb information, both to develop structure and to transform inputs into products that yield income (or, in the case of consumers, utility). It is the capacities of humans to seek out knowledge that can, in turn, exploit energy in creative ways, that differentiates complex economic systems from other kinds of complex systems. Stable meso populations are fundamental in such systems since they provide the essential order upon which complexity (or variety) can grow. It follows that the most coherent feature that we observe in a healthy economic structure is that its order is, to a significant extent, irreversible across a wide range of environmental conditions.⁵ And, equally, the periphery of such systems can be characterized by a bewildering amount of variety and change. Such systems survive and evolve through the generation of new variety and the avoidance of tendencies towards homogeneity (entropy and its informational analogue). So, microeconomic behaviour cannot be assumed to be homogenous simply because no evolution can occur in the presence of homogeneity, only system death in a world in which the second law of thermodynamics operates.

The heterogeneity that we observe in real systems is due to the presence of a myriad of idiosyncratic ways in which specific agents apply meso rules. A meso rule may be widely adopted in a population with high fidelity and efficacy, but the environments faced by adopters may vary considerably. This results in micro variety that, in addition to providing a very heterogeneous set of goods and services, can yield meso rule adaptations through processes of learning and selection. However, it is clear that theorizing about the emergence of new and adapted generic rules and change in meso populations cannot involve formal analysis, nor can the resultant economic outcomes be described in terms of analytical solutions. This would require structure to be invariant, i.e., all chosen elements and connections must be fixed. As we have noted, this has given rise to a simulation/calibration methodology in evolutionary economics that allows us to study how heterogeneous agents apply generic rules and shift from one generic rule to another when circumstances dictate that this is worthwhile.

However, once we think in terms of the micro-meso-macro framework, we cannot restrict our simulations to processes of selection (generally

⁵ In ecology, this is referred to as 'resilience'.

represented by replicator dynamics). It is also necessary to explore the self-organizational processes of learning by doing and incremental innovation as the meso population of rule carriers increases. Furthermore, the consequent flows of microeconomic value yield aggregate expenditure flows at the macroeconomic level and, in turn, these flows feed back into the decisions of individuals. So we have a two-way value flow interaction between the microeconomic and the macroeconomic that has, at its base, an interconnected set of meso rules.

3 A Methodology

So how can we design a methodology that is consistent with the micro-meso-macro framework? What we shall argue is that simulation/calibration is not, by itself, an adequate methodology to provide explanations of economic phenomena although it has an important role to play. In the absence of access to controlled experimentation (as must be the case in macroeconomics), the explanation of movements of economic data over time requires a methodology that is aimed at the discovery, not of the detailed specifics, for example, of firm behaviour, but of general principles that operate reliably for considerable periods of time. We shall look first at the meso, then the macro and, finally, the favoured starting point of many economists: the micro.⁶

3.1 The meso

Having identified a research question, the first step should be to examine the history of the economic entity and its components over the chosen time period of the investigation.⁷ The goal is to identify the different kinds of generic rules that enable value-generating connections between the components of identifiable systems. Case studies may also be necessary to augment historical accounts, particularly at the firm or industry level. The most easily observed generic rules are laws. However, it is often tacit norms and conventions that are crucial. We know from the outset that there are always localized and specific generic rules with small carrier populations – these are diverse across the economy and generate value in a manner that can look stochastic in aggregated data. The order and associated continuity of structure that we observe is due to the adoption and use of a set of widely held generic rules as stable meso units. However, we also know that meso rule populations grow and decline and, with them, the set of heterogeneous outputs.

⁶ This is obviously true of neoclassical economists but it is also true of many neo-Schumpeterian evolutionary economists who devote too little attention to the institutional context (the meso) and relatively little attention to Keynesian features of the economic system as a whole (the macro).

⁷ This is likely to be data determined.

Economic growth is the product of observed diffusion processes that involve meso rule connections (order) spreading across a population. A myriad of local modifications, generated by learning by doing and incremental innovation, yield heterogeneity in productive processes and products. Economic decline involves the onset of disconnections (disorder) in systems where the capacity to generate new processes and/or products has been exhausted. Unlike the diffusion process, this is unlikely to be smooth because meso structures built upon generic rules cannot be easily reversed and may have to be abandoned. It is also for this reason that structures that have reached their growth limit are often rapidly absorbed by other meso rule carrying structures that are more efficient or adaptable, through take-over and merger. Industry shakeouts are common (Klepper 1996), as is the demise of a short-lived fashion good. Careful historical study can enable us to understand these population dynamics and how populations interconnect. We need to know which generic rules form stable meso populations and which are subject to significant population change over a period of investigation.⁸

A precursor to this kind of meso rule perspective can be found in Foster's (1992) econometric study concerning the determination of British monetary magnitudes. It was found that what we would now identify as two meso rules seemed to be fundamental in the monetary process under investigation: one was the prudential regulatory and associated lender of last resort arrangements managed by the central bank and another was the conventional rules adopted by banks in their lending. The chosen explanatory model was built upon the observation that these rules seemed to be quite stable over the historical period in question, although, in the econometrics, allowance was made for some drift in the application of these rules. Once the core meso configuration of rules had been identified, implied relationships between aggregate time series variables were identified. To this core were added hypotheses concerning the role of 'price' variables of the kind suggested by conventional economic theories. Importantly, the inclusion of such variables was justified by the observation that a workable market existed, i.e., a meso rule concerning trade – in this case in a well-regulated short-term money market. The resultant 'evolutionary macroeconomic' model proved to be much superior to conventional demand for money based models both in explanation and prediction.⁹ Foster and Wild (1999a) went on to take a similar approach to modeling monetary (non-bank) time series data emanating from an observable diffusion process, following a regulatory (meso rule) change. Again, this

⁸ Care has to be taken in such investigations not to confuse fads and fashions (short-lived meso rules) with core meso rules that are pervasive over the period in question and likely to continue to be so beyond.

⁹ Further historical investigations revealed that, because of colonial and post-colonial connections, Australians had adopted very similar meso rules in their banking system and, in Foster (1994), remarkably similar econometric results were found, using a model that was only adapted in relatively minor ways to account for what were mainly demographic and property market differences (due to climatic and geographical variations).

approach was found to be significantly more successful than one based on conventional theory and methods and offered evidence of much more clearly identified 'price incentive' effects.

Although these studies involved models constructed from meso rules, identified by historical study, a full micro-meso-macro methodology was not adopted. In particular, the heterogeneity of microeconomic behaviour and the associated adaptation of generic rules for local application (adding variety to the meso population) were not addressed. So, although good explanatory econometric models were obtained, no understanding of the underlying process, whereby heterogeneous economic agents adopt and adapt generic rules to generate stocks and flows of economic value, was obtained. From the perspective here, a 'meso-macro' approach was adopted. This was an important deficiency because it permitted conventional economists to conclude that the chosen methodology did not contribute to theoretical understanding, beyond some add-ons drawn from conventional theorizing. However, before we address this matter properly, the macro dimension of the methodology that we are suggesting requires discussion and development.

3.2 The macro

Aggregate time series data provide useful information about key connections in the economic system. They offer a particular kind of history. For example: aggregations of value, in income or expenditure form, flowing from heterogeneous economic processes; aggregations of the value of economic structures, as measured in asset and liability valuations; series of prices that are often weighted averages of many price indices; aggregations of inputs and output quantities, often computed using price deflators; aggregations of qualitative factors, such as numbers of patents or indices of consumer sentiment. It is well known that there are serious aggregation problems in connecting theoretical representations of heterogeneous microeconomic behaviour with macroeconomic aggregates. As noted, the conventional solution in modern macroeconomics is just to pretend that all microeconomic behaviour is homogenous and deal only with a 'representative agent' operating under very strong and unrealistic rules, some of which are relaxed in thought experiments. Essentially, what such a methodology involves is a theoretical starting point that is a fully connected network of identical sets of components across all space and all time, and, moreover, one in which the meso domain of rules is given and invariant. A universal rule – the constrained optimization approach – is employed to ensure that an equilibrium outcome can be defined on the basis of logic.

However, such a methodology is not concerned with the actual processes that underlie the historical data under consideration, nor is the architecture of the structures that enable such processes to occur of any direct interest. In the real world, economic structures are networks of components with sparse connections that can increase or decrease in number. Connections can be

local or long distance, involving only a few or many elements. For (dissipative) structures to survive, they must operate within a set of meso rules. But economic structures must also have operational micro rules to deal with specific aspects of production processes and product characteristics. Thus, we can think of the value that is generated at the heterogeneous microeconomic level as being separable into two components at some defined level of aggregation and defined time period: (1) there is the 'order' effect of widely held generic rules forming meso units, and (2) the 'complexity' effect of system specific operational micro rules. The existence of the former will show up in aggregate value data, both in autoregressive tendencies, because of the persistence of meso rules, and in enduring statistical associations between aggregate variables. The latter involves operational micro rules that are connected hierarchically with generic meso rules but are very heterogeneous and disconnected on the periphery of systems. The value generated on the periphery is subject to constant change, as incremental innovations and learning by doing results in novel micro rules that are embodied in new products and processes while old ones are discontinued. This is reflected in statistical noise in aggregate value data. However, components (1) and (2) are connected because some micro rules become new meso rules in the form of new institutions while old meso rules are abandoned. Equally, micro rules depend, critically, upon the existence of stable populations of meso rules. These micro-meso interactions are the source important shifts in aggregate associations and fluctuations in autoregressive parameters.

So, underlying time series relating to processes and outcomes in particular economic structures is a mix of generic (meso) and idiosyncratic (micro) rules that produce a mixture of auto-regressive tendencies and fluctuations in aggregate data. And this is what we typically see: we observe time series following paths that, superficially, are like either random walks or random walks with drift (Ormerod 2005). As is well known, series with unit roots are historical in nature and cannot be viewed as deviations from some equilibrium value. The frequency with which they occur suggest that we should take a historical approach to understanding them (Freeman and Louçã 2001). However, conventional econometric modellers do not do this but instead, seek to remove the autoregressive component by 'first differencing' it to render a series 'stationary' in order to focus upon the associations between time series. In doing this, they eliminate a key connection between the data and the meso rules that are the source of economic value, as well as key variations in the two way relationship between meso and micro rules.

As Foster and Wild (1999b) point out, this involves the removal of very important information concerning the role of meso rules in maintaining and growing economic value over time. In particular, the impact of the spread of new generic rules, through incremental innovation and learning by doing in an increasing population of rule carriers, is ignored despite the widespread observation of value trajectories following logistic diffusion curves. Furthermore, structural discontinuities in time series models are 'patched up' by various *ad*

hoc ‘regime shift,’ ‘break-point’ and ‘regression switching’ techniques rather than investigated through proper historical inquiry.

Lets us think about aggregate data and underlying economic processes from an evolutionary perspective. Value flows registering in macroeconomic data emanate from processes in dissipative, knowledge rich structures in the following simple way:

$$X = X_{-1} + X_N - X_S \quad (1)$$

Where

X_N : new value from the application of new meso rules and/or increased use of existing meso rules

X_S : loss of value because of the abandonment of old meso rules and/or the reduced use of prevailing meso rules

X_N and X_S cannot be zero, because it would be an untenable state for any dissipative structure to be in because of the second law of thermodynamics. $X_N = X_S$ is also untenable as an enduring state because the nonlinear nature of entropy processes ensures that it is a structurally unstable state, i.e., eventually X_S will exceed X_N . If X_N and X_S are random, and $X_{-1} > 0$ then we get a classic random walk.¹⁰ However, in evolving economic structures, X_N and X_S are not random, they are the outcome of managerial and/or entrepreneurial decision-making (or lack thereof). Following the entropy law, X_S has a component that is related to X_{-1} because of component and connective wear out, for example, because of the breakdown of technological rules embodied in physical assets. Firms try to anticipate this in their depreciation provisions. This results in planned replacement investment. However, this need not just be a negation of X_S but also an addition to X_N , inasmuch as replacement capital goods embody new technological rules.

The biggest problem is the loss of *relevance* of productive structure – both technological and organizational rule obsolescence lead to unanticipated rises in X_S which can be fatal in an economic structure which is not generating enough X_N through the application of new meso rules. Thus, new investments in capital goods, human capital and organizational structures always have to have an element of entrepreneurship if a firm is to survive. So, although some of the decline in old value and the increase in new value are linked to X_{-1} , some value change is uncertain – there is unplanned obsolescence of existing rules and the emergence of new structures embodying entirely new rules:

$$X = X_{-1} + nX_{-1} - sX_{-1} + u \quad (2)$$

or

$$X = X_{-1}(1 + n - s) + u \quad (3)$$

¹⁰ The random walk hypothesis is often supported in time series data over certain periods, particularly in the case of financial asset prices.

Where $\mathbf{u}$ is the combined value associated with X_N and X_S that is unconnected with embodied meso rules in existing structure.

If $\mathbf{u}$ happens to be random (often presumed to be a reasonable assumption in evolutionary biology) and n is not equal to s , we have a random walk with ‘density dependent’ drift. For growth to occur, n must be greater than s . But, this growth cannot go on forever – we observe a ‘self-organisational’ rise in value that tends to a limit often logistic in form. This implies that n and s are not fixed parameters – density dependent growth occurs because of incremental innovation and learning by doing which must, eventually, run out if there is not a shift to the application of a radically different meso rule. For a logistic path to exist $n - s = b(1 - X_{-1}/K)$, where b is the logistic diffusion parameter. However, this is limited to growth up to the point where $n = s$ or, equivalently, where $X_{-1} = K$. If conditions arise where worn out structure is not being fully replaced ($n < s$) then a downward spiral of decline will set in. So, whereas $n > s$ is a state where there is full replacement of worn out structure plus incremental innovation/learning, $n < s$ is one where exponential decline is inevitable.

Once we account for the incremental application of meso rules, i.e., self-organisation, $\mathbf{u}$ is likely to be more random in character. However, over the time span of a logistic curve, $\mathbf{u}$ is still likely to be non-random because the arrival and departure of meso rules and their applications is not random. The adoption of new meso rules in favour of old ones is most easily enacted around the point of inflection on the logistic curve when value is growing fastest and surpluses for investment tend to be largest. In contrast, it becomes very difficult to shift meso rules (as a stable population, or institution) at a growth limit. Thus, the variance of $\mathbf{u}$ is not likely to remain constant, either in its oscillatory characteristics or in its variance (Foster and Wild 1999b), ie, we cannot presume that $\mathbf{u}$ is Gaussian.

So we can see that there are interesting features to be discovered in aggregated data, given the connections that exist between meso rules and value generation. Understanding the statistical properties of relevant macro time series data and the associations between them can, therefore, provide important information to compare with prior findings concerning meso rules obtained by historical and case study. We can identify if time series over a particular period are stationary or non-stationary and whether a trend is deterministic (linear or nonlinear) or stochastic (or both) with or without time drift using conventional methods. From our standpoint here, deterministic trends are indicative of the operation of meso rules in generating historical path dependence. In addition, co-integration tests can establish the degree of connectivity between time series which, again from the standpoint here, would suggest that there exist network connections due to adherence to meso rules. For example, we may observe path dependence in a measure of output, perhaps a logistic path, and we may also find co-integration with an input measure. Conventional economists would tend to see this as indicating the existence of a ‘long run equilibrium’ production function. Shifts of

this presumed function might then be attributed to exogenous technological change. From our perspective this co-integration can be viewed as evidence of meso rules in operation, with a non-stationary, non-equilibrium diffusion process of endogenous technological and organizational change.

A statistical examination of the properties of available and relevant series and the connections between them can yield information about the possible existence of generic rules forming meso units as well as variations in the populations of rule adherents. Once these statistical associations have been identified, qualitative evidence, already compiled through historical and/or case studies can be compared. We can then proceed to eliminate meso rules that seem to have no counterpart in the time series data that we have at our disposal and we can also eliminate time series and connections between time series that have no obvious link with identified meso rules. A 'parsimonious' model can then be constructed and estimated econometrically. What we then have are estimated parameters that reflect the connections between time series consistent with the operation of underlying meso rules. It should be stressed that this is not just 'sociology' – in economic investigations the most important meso rules are those, for example, concerning the collective acceptance of market structures and adherence to formal and informal contractual obligations. This means that quite conventional relationships involving prices, incomes and quantities are likely to play a significant role.

3.3 The micro

The problem with the parsimonious approach outlined is that it cannot span phases of rapid transition when meso rules rapidly lose their populations. Regression is, essentially, an averaging technique that cannot easily encompass non-average behaviour. This problem was encountered in Foster and Wild (1999a) and investigated through a residual error decomposition approach in Foster and Wild (1999b). Evidence was produced that the oscillatory behaviour of residuals and the variance of these residuals revealed certain patterns prior to structural shifts (in the case in question, because of regulatory changes). This seemed to be useful because it suggested that it is possible to get an 'early warning' that a system is due to stop growing and enter a phase of structural instability. However, by necessity, the microeconomic behaviour that gave rise to such transitions is left firmly in a 'black box'. It is for this reason that conventional economists tend to look at this kind of research and argue that it lacks theoretical micro-foundations. And they have a point because, although the meso remains the driver of microeconomic behaviour and the generation of value, ultimately, we need to know about the motivations that give rise products and processes at the microeconomic level and how these motivations eventually result in structural transitions.

It is here that simulation/calibration is very useful since it is immediately clear that, in the evolutionary setting that we are envisaging, there is no possibility of finding analytical solutions in the traditional way. But we can

construct simulations of how agents use meso rules to devise heterogeneous ways of generating economic value. Calibration with historical data is not an empirical exercise but, rather, one of seeking out theoretical propositions concerning the economic processes at work.

The first step is to use historical study to identify the appropriate definitions of the agents involved, following, for example, Malerba et al. (2001). However, unlike in the latter study, this methodology does not seek specific details but, rather, general characterizations of economic agents and how they behave and interact with each other. There is no reason why the logic of neo-classical microeconomics and game theory should not be used here, provided that it is appropriate to the environment faced by decision-makers. It is essential that the characterization of behaviour chosen is realistic within the meso-rule structure that has been found to exist and has a counterpart in the data.

The main difficulty faced by all simulator/calibrators, is to justify the parameters used in simulations. Critics argue that the judicious selection of parameters can allow many different simulations to calibrate on the same data. In the methodology proposed here, selection is restricted. The chosen parameters must, ultimately, be consistent with the econometrically estimated parameters connecting relevant time series variables. So calibration is not just on the time series data but on the statistical associations between time series, reflecting the existence of significant meso rule populations. In other words, when we add up the value of our simulated agents, in line with real data definitions, we should see the same parametric relationships. This is a much stronger than usual test. But it remains limited because we know that estimated econometric parameters only make sense when there are no structural discontinuities. So, the most stringent test of a simulation is to calibrate it beyond an estimation period where there is no discontinuity into one where there is. Such a test is based upon the presumption that the seeds of discontinuity are present, even when value is growing in a stable way. In other words, there must be aspects of behaviour in the good times that contribute to the emergence of bad times.¹¹ So this methodology investigates non-average behaviour through simulation by using evidence of average behaviour offered by econometric modelling.

What such simulations can achieve is an analytical understanding of non-linear microeconomic behaviour in the face of a set of meso rules. In a sense, it can be classified as inductive theorizing but 'abductive' (see Josephson and Josephson 1994) is more appropriate because the econometric modelling involved uses some deductive reasoning concerning the properties of dissipative structures and some traditional deductive reasoning. The agent based simulation design is also likely to involve some logical reasoning, i.e. some appeal to rational behaviour, however incomplete or bounded, is likely to be applied. The goal of simulation/calibration of this kind is theoretical: the quest

¹¹ For example, if a firm distributes too much profit and under-invests when growth is high, it will, undoubtedly run into difficulties if its market saturates (Foster 1986).

is to discover recurring theoretical principles operating in a range of different circumstances using a methodology that is firmly connected to the identification of rules in history and statistical relations in historical data.

In conventional economics, fluctuating time series are frequently modelled analytically using dynamic mathematics. Indeed, this approach has almost become the core of modern macroeconomics. However, it is a fundamentally mistaken theoretical approach because time series data do not actually measure behavioural processes but instead are only an aggregated value manifestation of such processes that are inherently complicated. What the micro-meso-macro methodology offers is a way of obtaining simplified, i.e., truly theoretical, representations of actual complex processes by taking a four level approach: historical/case study; time series statistical investigation, parsimonious econometric modelling, micro-simulation/macro-calibration.

This is markedly different to conventional modelling in three ways: first, non-stationarity in time series, which conventional modellers desire to eliminate in order to focus upon supposed 'equilibrium' relationships, is viewed as crucially important and reflective of the meso rule structure that drives economic activity; second, *heterogeneous* microeconomic behaviour is dealt with explicitly and theoretically; third, the methodology embraces uncertainty, path-dependence and the success and failures that characterise economic evolution - there is no escape into a non-existent world of abstraction yet, at the same time, there is no appeal to 'irrationality' in the choices that people make.

4 Conclusion

In this paper, we have offered a methodology for evolutionary economic analysis that seeks to combine computational, historical and econometric methodologies into an integrated methodology built upon the micro-meso-macro framework. We think that this approach is superior to simulation/calibration approaches in new classical macroeconomics, in the dynamical economic approaches to be found in neoclassical, and post-Keynesian and new Keynesian analysis, and also in the historical/case study approach of institutional economics. To a large degree, our methodology encompasses what these approaches seek to do in a unified framework. Simulation and calibration are therefore not ends in themselves, but useful items in an analytical toolkit that can help us understand the structures and processes that we find in economic reality. We think that economic structure is both ontologically and analytically best represented as a complex system of generic rules forming meso units and that the methodology discussed reflects this. And, as we have argued, this involves a hybrid methodology of historical investigations, case studies, statistical analysis, econometric modelling and simulation/calibration. The economy is a complex adaptive system and so we should not be surprised that its features cannot be adequately revealed with the prevailing 'simplistic' methodology.

In moving towards such a methodology we are, of course, shadowing the insights of others. The notion of meso rules and their emergence and decline owes much to Hayek and his notion of spontaneous order, the diffusing character of meso rule adoption and application is closely connected with Schumpeter's insights and the unique role of knowledge in economics goes back to Marshall's vision of how the economic system works. Also, Hayek wrote about patterns of rules and their enduring qualities. In this regard, the methodology that is proposed offers potential as a basis for mapping the generic structure of the economy.

Aside from more exotic quests of this kind, we believe that the proposed methodology should be of interest and relevance beyond evolutionary economics. We know that difficulties in relating macroeconomics to microeconomics have posed serious problems in modern macroeconomics and that these problems are, at base, methodological in character. But it is also the case that evolutionary economists have relatively little to say about macroeconomics. This is commonly justified on the ground that aggregation averages out all the non-average variety that is the fuel of economic evolution so that there is little of interest left. However, the complex systems approach challenges this presumption on the ground that we must understand where variety comes from (Foster and Metcalfe 2001). We argue that the evolutionary economic process is driven from the meso, even though micro agents are the source of all ideas (and, therefore the rules that may then form into meso rules) and the domain of all economic actions and operations. If this is accepted then it must be made explicit in the methodologies that we use.

References

- Altug S. (1989) Time to build and aggregate fluctuations: some new evidence. *Int Econ Rev* 30:889–920
- Brenner T, Werker C (2006) A practical guide to inference in simulation models. *Papers on economics and evolution* #0602, Max Plank Institute of Economics, Jena
- Dopfer K, Foster J, Potts J (2004) Micro meso macro. *J Evol Econ* 14:263–279
- Dopfer K, Potts J (2004) Micro–meso–macro: a new framework for evolutionary economic analysis. In: Metcalfe JS, Foster J (eds) *Evolution and economic complexity*. Cheltenham, Edward Elgar
- Dopfer K, Potts J (2007) *The general theory of economic evolution*. Routledge, London
- Earl P, Potts J (2004) The market for preferences. *Cambridge J Econ* 28:619–633
- Foster R (1986) *Innovation: the attackers advantage*. Summir Books, New York
- Foster J (1992) The determination of Sterling M3, 1963–88: an evolutionary macroeconomic approach. *Econ J* 102:481–496
- Foster J (1994) An evolutionary macroeconomic model of Australian dollar M3 determination: 1967–93. *Appl Econ* 26:1109–1120
- Foster J (1997) The analytical foundations of evolutionary economics: from biological analogy to economic self-organisation. *Struct Change Econ Dynam* 8:427–451
- Foster J (2000) Competitive selection, self organization and Joseph A. Schumpeter. *J Evolution Econ* 10:311–328
- Foster J (2005) From simplistic to complex adaptive systems in economics. *Cambridge J Econ* 29:873–892

- Foster J, Metcalfe JS (eds) (2001) *Frontiers of evolutionary economics*. Edward Elgar, Cheltenham
- Foster J, Potts J (2006) Complexity, networks and the importance of demand and consumption in economic evolution. In: McKelvey M, Holman F (eds) *Flexibility and stability in economic transformation*. Oxford University Press, Oxford
- Foster J, Wild P, (1999a) Econometric modelling in the presence of evolutionary change. *Cambridge J Econ* 23:749–770
- Foster J, Wild P (1999b) Detecting self-organisational change in economic processes exhibiting logistic growth. *J Evolution Econ* 9:109–133
- Freeman C, Louçã F (2001) *As time goes by: from the industrial revolutions to the information revolution*. Oxford University Press, Oxford
- Hoover K (1995) Facts and artifacts: calibration and the empirical assessment of real-business cycle models. *Oxford Econ Pap* 47:24–44
- Josephson J, Josephson S (eds) (1994) *Abductive inference: computation, philosophy, technology*. Cambridge University Press, Cambridge
- Kirman A (1983) Communication in markets: a suggested approach. *Econ Lett* 12:101–108
- Klepper S (1996) Exit, entry, growth and innovation over the product life cycle. *Am Econ Rev* 86:562–583
- Klochko MA, Ordeshook PC (2006) *Endogenous time preferences in social networks*. Edward Elgar, Cheltenham
- Kydland F, Prescott E (1982) Time to build and aggregate fluctuations. *Econometrica* 50:1345–1370
- Kydland F, Prescott E (1996) The computational experiment: an econometric tool. *J Econ Perspect* 10:69–85
- Malerba F, Nelson R, Orsenigo L, Winter S (2001) History-friendly models: an overview of the case of the computer. *J Artif Soc & Soc Simul* 4(3):6
- Metcalfe JS, Foster J, Ramlogan R (2006) Adaptive economic growth. *Cambridge J Econ* 30:7–32
- Nelson R, Winter S (1982) *An evolutionary theory of economic change*. Harvard University Press, Cambridge
- Ormerod P (2005) *Why must things fail: evolution, extinction and economics*. Faber & Faber, London
- Potts J (2000) *The new evolutionary microeconomics: complexity, competence and adaptive behaviour*. Edward Elgar, Cheltenham
- Werker C, Brenner T (2004) Empirical calibration of simulation models. *Papers on economics and evolution* #0410, Max Plank Institute of Economics, Jena
- Witt U (1997) Self-organization and economics – What is new? *Struct Change Econ Dyn* 8:489–507